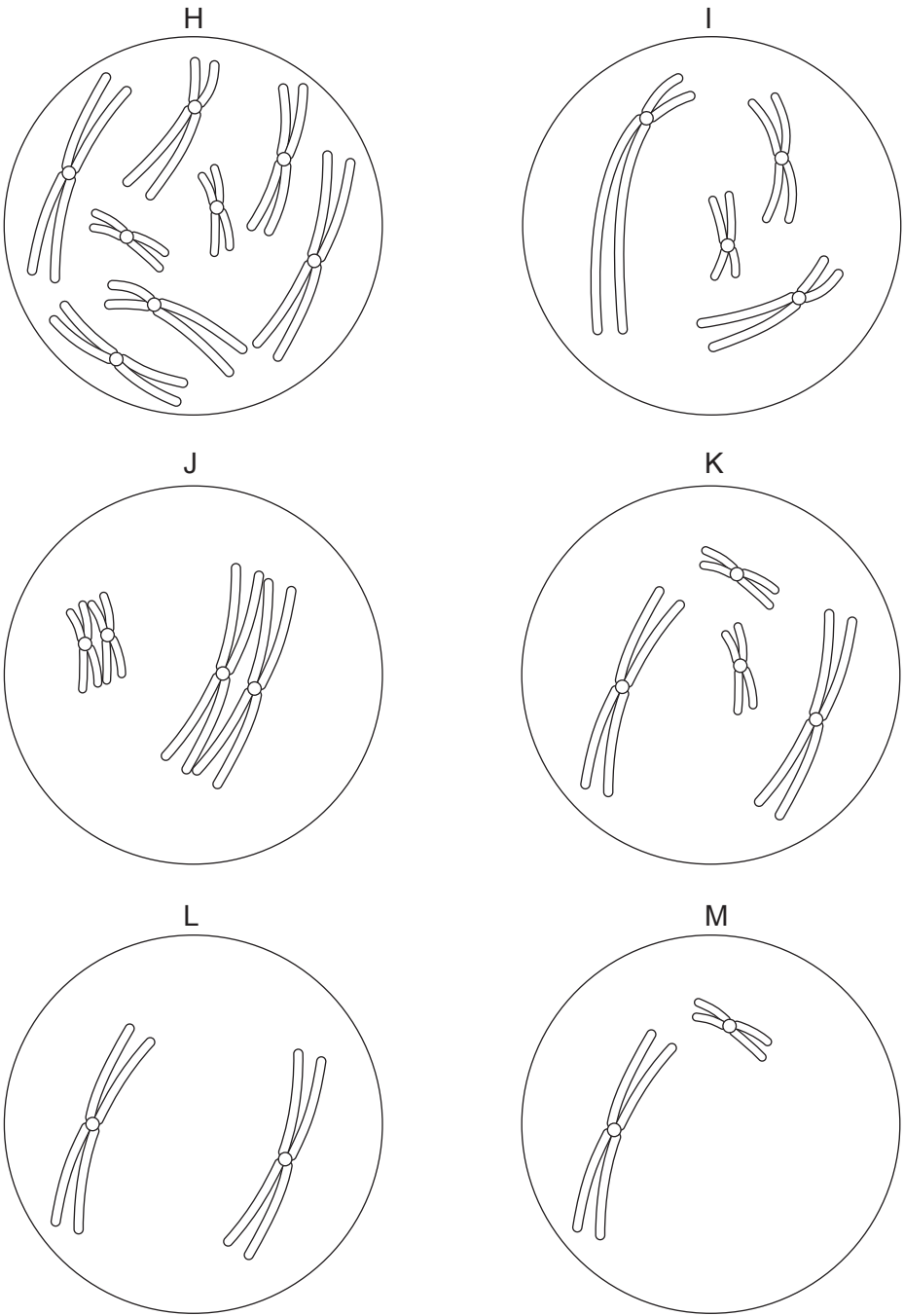


3. Spider mites are small invertebrates which are related to spiders and scorpions. One species, *Eutetranychus africanus*, which is native to the island of Mauritius, has very few chromosomes, $2n = 4$.

The diagrams below show the chromosomes in some cells undergoing mitosis or meiosis.



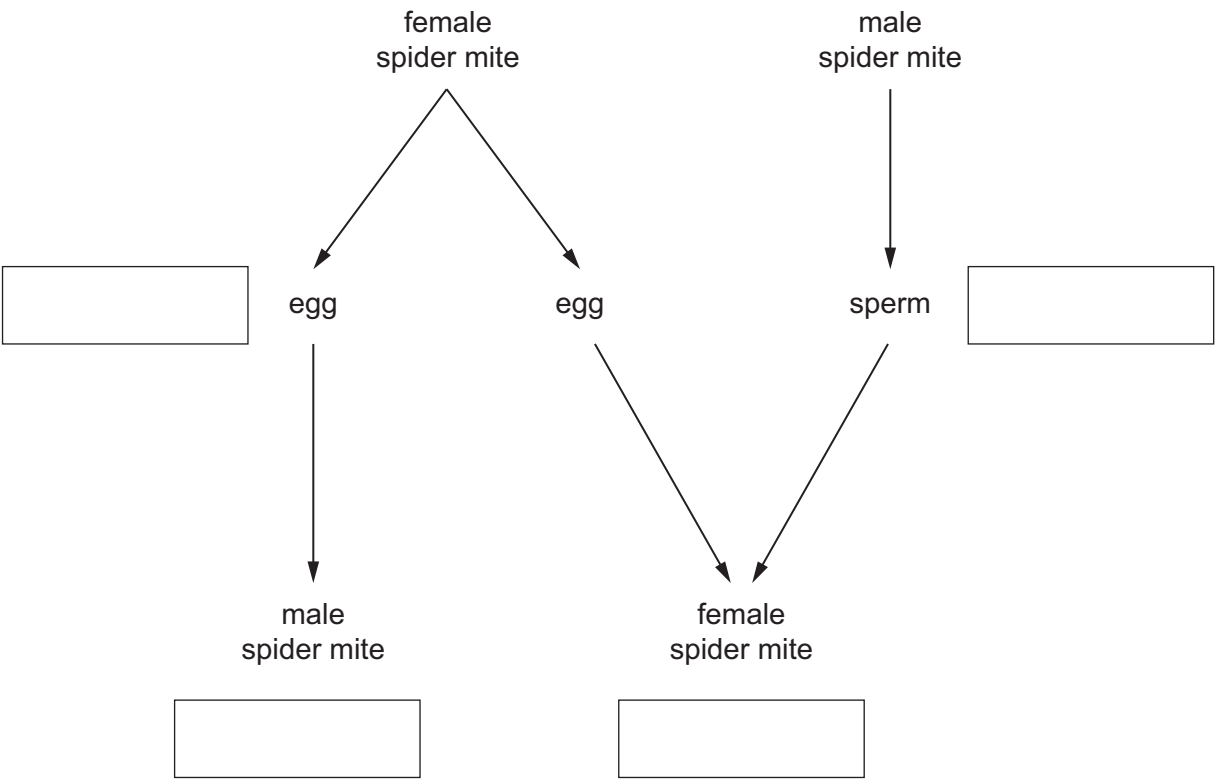
(a) Identify the cells that belong to this species of spider mite.

[2]

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(b) The diagram below shows the individuals that make up a colony of spider mites. Female spider mites are produced when a male sperm fertilises a female's egg. Male spider mites develop from an unfertilised female egg.



- (i) **Complete** the diagram to state which stages are **haploid** or **diploid**. [2]
- (ii) What conclusions can be made about the formation of male and female gametes in spider mites? [2]

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Examiner
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- (c) (i) A sample of tissue from the **leg muscle** of a young spider mite was analysed. The mass of DNA in some of the cells was 6.8 arbitrary units (au), whilst in other cells it was 3.4 au. Suggest explanations for this difference using your knowledge of the cell cycle. [3]

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- (ii) The table below shows the percentage of cells, with either 3.4 au or 6.8 au of DNA, in the leg muscle of a young spider mite and an older spider mite.

Mass of DNA in cell /au	Percentage of cells	
	Young spider mite	Older spider mite
6.8	20	5
3.4	80	95

What conclusions can be drawn from the data regarding the significance of mitosis in these spider mites? [2]

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